SANDIP UNIVERSITY (UGC Recognised) NAAC GRADE A

School of Computer Sciences and Engineering
Department of Computer Science and Application

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A PROJECT REPORT

ON

"AI-Driven Predictive Analytics on Student Academic Performance and Tool Usage Patterns"

UNDER THE FACULTY OF COMPUTER SCIENCE & APPLICATION

Submitted by

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CERTIFICATE

This is to certify that **Ms. Mahima Chaudhari**, PRN No**. 250105051019** Student of MCA Semester-II has successfully completed the Project work on Title **" AI-Driven Predictive Analytics on Student Academic Performance and Tool Usage Patterns"** under my guidance. This report submitted to Department of Computer Science and Application, School of Computer Sciences and Engineering, Sandip University, Nashik in the AY 2026-27 for partial fulfillment and requirement for the End Semester Examination.

Prof. Smita Dhawadekar
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Examiner 1 **Examiner 2**

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DECLARATION

I hereby declare that the Project work titled **"AI-Driven Predictive Analytics on Student Academic Performance and Tool Usage Patterns"** submitted to Sandip University, Nashik is a record of an original work done by me under the guidance of **Prof. Smita Dhawadekar** Department of Computer Science and Application, School of Computer Sciences and Engineering, Sandip University in the AY 2026-27 for a partial fulfillment of requirement for the Completion of End Semester Examination. This report has not been submitted to any other University or Institute for the award of any degree or diploma.

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Abstract

The rapid integration of generative Artificial Intelligence (AI) platforms in higher education has fundamentally transformed student learning patterns, academic research workflows, and problem-solving methodologies. While tools such as ChatGPT, Google Gemini, and GitHub Copilot offer substantial efficiency gains, their quantifiable impact on academic performance and final grade outcomes remains a critical area of study. Traditional educational evaluation frameworks rely primarily on descriptive, static surveys that fail to model non-linear interactions between multi-variable parameters such as usage duration, institutional policies, infrastructure access, and discipline-specific demands.

To address this gap, this project presents an end-to-end data intelligence and predictive analytics pipeline that integrates Deep Learning with Business Intelligence (BI) visualization. Using a structured dataset of 3,614 student records across technical, managerial, and humanities disciplines, the system implements automated feature preprocessing using Scikit-Learn's ColumnTransformer, standardizing continuous variables via StandardScaler and encoding categorical variables using OneHotEncoder. A 4-layer TensorFlow/Keras Sequential Neural Network ($64 \rightarrow 32 \rightarrow 16 \rightarrow 1$ nodes) is trained using ReLU activations and the Adam optimizer to predict student grade impact scores (Predicted_Impact_on_Grades). Prediction error metrics (Absolute Error $= |\text{Actual} - \text{Predicted}|$) are computed, achieving a high model accuracy with an overall Mean Absolute Error (MAE) of 1.25 points.

Finally, the processed predictive outputs are exported to Microsoft Power BI Desktop to construct an interactive, multi-dimensional executive dashboard. The visual interface features KPI callout cards, stream comparison clustered bar charts, usage-hour line trends, tool adoption donut charts, and dynamic cohort slicers. This integrated architecture successfully bridges machine learning regression with executive visual analytics, providing educational administrators and policy makers with data-driven decision support.

Keywords: Artificial Intelligence in Education, Predictive Analytics, Educational Data Mining (EDM), TensorFlow, Deep Neural Networks, Machine Learning Regression, Feature Engineering, Microsoft Power BI, Data Visual Analytics, Student Academic Performance.

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